

GAURAV GANDHI

Senior Data Scientist · GenAI & LLM Applications · RAG & Agentic Systems · Fine-Tuning · 5 Years

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PROFESSIONAL SUMMARY

Senior Data Scientist with 5 years building and shipping enterprise GenAI and machine-learning products end-to-end — from business problem through production deployment, monitoring, and iteration. Now lead and mentor a 5-person team on Uber's AI org, delivering document-intelligence systems, multi-agent RAG copilots, and LLM fine-tuning pipelines in production. Strong modeling depth across the LLM and deep-learning stack: transformer and diffusion architectures, two-tower / contrastive retrieval, parameter-efficient fine-tuning (LoRA/QLoRA), distributed training (PyTorch, Ray, DeepSpeed), and rigorous evaluation — with the production engineering (Python, FastAPI, GCP) to ship models into live systems.

PROFESSIONAL EXPERIENCE

Indium Software (Client: Uber Technologies — Uber AI)

Jun 2024 – Present

Senior Data Scientist

May 2025 – Present · Bengaluru, India

- Lead and mentor a 5-person data-science team across GenAI document-intelligence and conversational-AI workstreams — owning modeling roadmap, evaluation strategy, and technical review.
- **LLM Fine-Tuning for Document Intelligence (LoRA / QLoRA)**: Raised document-extraction field-level accuracy to 97%+ by building a parameter-efficient fine-tuning framework (Llama 3.1, GPT-Neo) with a layer-freezing strategy, instruction-tuning data design, and an automated field-accuracy eval harness; mentored multiple teams on prompt and evaluator design.
- **Vision Document-Quality Gate for Auto-Approval / Rejection (ViT)**: Removed the manual-review bottleneck in earner document verification by training a Vision Transformer that scores each upload against 9 rejection reasons (blur, glare/shadow, missing corners, damaged, irrelevant, occluded, and more) over 55 document types, auto-approving only below a calibrated per-(type, reason) human-recall threshold.
- **Multi-Agent GenAI Copilot for Enterprise Data (Hybrid RAG + NL2SQL)**: Cut analytics turnaround from hours to seconds for non-technical teams by designing a production multi-agent copilot combining Hybrid RAG (dense + sparse retrieval), NL2SQL, a schema knowledge graph, and LangChain tool orchestration — live across business units.

Data Scientist

Jun 2024 – May 2025 · Bengaluru, India

- **Document-Understanding Transformer at Production Scale**: Owned the multi-task training objective and extraction head for an encoder-decoder Transformer pretrained on 50M+ documents across 144 A100 GPUs (PyTorch, Ray Train, DeepSpeed ZeRO-3) — shipped to production at 95%+ field accuracy, displacing manual review for \$10M+ in annual cost savings.
- **Session-Aware Recommendation Engine (BERT + Sequence Modelling)**: Drove a sustained 27% lift in user engagement by replacing static collaborative filtering with a BERT-based session-aware sequence model capturing evolving user intent, validated via A/B tests across releases.

FedEx Express (Consumer Logistics & Supply Chain)

Aug 2022 – Jun 2024

Decision Scientist

Bengaluru, India (Remote)

- **Probabilistic Anomaly Detection for Account Retention**: Recovered lost enterprise shipping volume — surfaced via Average-Daily-Volume decline — by deploying an ensemble of Bayesian change-point detection, probabilistic clustering, and profitability scoring, shaping regional supply strategy cross-functionally.
- **Demand Forecasting & Capacity Optimisation (SARIMA + Bayesian Tuning)**: Reduced operational inefficiencies ~20% by upgrading ARIMA pipelines to SARIMA with Box-Jenkins tuning, Fourier seasonality, and automated anomaly filtering, feeding sales and staffing plans.

Tata Consultancy Services (TCS)

Jul 2021 – Jul 2022

Data Engineer

Bengaluru, India (Remote)

- **High-Volume Data Pipelines & Analytics (GCP)**: Ensured reliable, scalable downstream analytics for a high-volume booking/transaction client by designing end-to-end GCP ETL pipelines plus dashboards for campaign targeting, segmentation, and trend monitoring.

APPLIED AI PROJECTS

- **Multi-Agent Flight Search** [\[GitHub\]](#) [\[Live Demo\]](#) — Turned natural-language trip requests into Pareto-ranked best-value and best-experience itineraries with multi-turn refinement and judged coherence, by building a Planner-to-Optimizer multi-agent system with a ConversationManager agent and an LLM-as-judge eval harness; live via SSE streaming to a Next.js UI.
- **TriagelQ — ML Issue-Triage System** [\[GitHub\]](#) [\[Live Demo\]](#) — Cut maintainer time spent sorting unlabelled issues, by building a four-stage triage service — a temperature-calibrated TF-IDF classifier, BGE + FAISS similar-issue retrieval, a LightGBM quantile estimator with calibrated prediction intervals (~77% coverage), and Groq-LLM synthesis into a validated triage plan.
- **Multimodal Fashion Recommender** [\[GitHub\]](#) [\[Live Demo\]](#) — Improved Recall@10 by 3.06x over a popularity baseline (2.12x over co-purchase) by training a two-tower recommender that fuses CLIP ViT-B/32 image embeddings with SBERT text embeddings using in-batch InfoNCE contrastive loss and FAISS retrieval, validated by modality ablation.
- **AetherArt — Diffusion Image Generation + Custom LoRA** [\[GitHub\]](#) [\[Live Demo\]](#) — Generated studio-quality images within an 8 GB consumer-GPU budget by training a custom rank-8 Ukiyo-e SDXL LoRA composed with Hyper-SD few-step distillation, adding ControlNet (Canny + Depth) conditioning, and selecting schedulers via a 360-run CLIP-score sweep; served on GCP Cloud Run (L4 GPU).
- **Agentic Fashion Shopping Assistant** [\[GitHub\]](#) — Enabled natural-language search over a 100k+ item catalogue without hallucinated product facts, by building a LangGraph six-tool agent over hybrid retrieval (dense FAISS + BM25 fused via Reciprocal Rank Fusion) with an LLM reranker and a sentence-level grounding layer that scrubs invented price/size/fabric claims.
- **ReviewIQ — LLM Review Intelligence** [\[GitHub\]](#) — Converted free-text reviews into structured fields plus an authenticity score across English, Hindi, and Hinglish, by building a tiered Groq-LLM extraction service (Llama 3.3 70B, JSON mode) with prompt-injection defense, PII redaction, multi-tenant support, and a CI-gated field-accuracy eval set.

KEY TECHNICAL SKILLS

- **Machine Learning & Deep Learning:** Transformer Architectures, BERT, RoBERTa, GPT, Llama, CLIP, ViT, Diffusion (SDXL), CNNs, LSTMs, Sequence Modelling, Two-Tower / Contrastive (InfoNCE), Recommender Systems, Feature Engineering, Model Evaluation, A/B Testing & Experimentation, Hypothesis Testing, Bayesian Inference, Time-Series
- **LLMs, GenAI & Fine-Tuning:** LLM APIs (OpenAI, Anthropic, Gemini, Hugging Face, Groq), Fine-Tuning (LoRA / QLoRA, PEFT, Instruction Tuning), Prompt Engineering & Evaluation (LLM-as-Judge), RLHF / DPO, Guardrails & Grounding
- **RAG, Retrieval & Agents:** RAG, Hybrid Retrieval (dense + BM25 / RRF), Vector Search (FAISS, pgvector, Pinecone), NL2SQL, Knowledge Graphs, Multi-Agent Systems, LangChain, LangGraph
- **Modeling Stack:** PyTorch, Hugging Face Transformers, diffusers, PEFT, sentence-transformers, scikit-learn, LightGBM, NumPy, Pandas, Polars, FastAPI
- **Programming:** Python (Advanced), SQL, Spark SQL, PySpark, Git, OOP, Unit & Integration Testing
- **Deployment & MLOps:** Docker, GCP (Cloud Run, Vertex AI), Azure, MLflow, GitHub Actions, model evaluation & monitoring; Ray Train, DeepSpeed ZeRO-3 for distributed training

EDUCATION

Post Graduate Diploma in Data Science — **Great Lakes Institute of Management** Feb 2022 – Nov 2022
GPA: 8.45 / 10.0 | Chennai, India (Remote)

B.Tech, Electronics & Communication Engineering — **Manipal University Jaipur** Jul 2017 – May 2021
GPA: 8.05 / 10.0 | Jaipur, India

CERTIFICATIONS

NLP Specialization · Deep Learning Specialization · LLMOps Specialization (Andrew Ng, DeepLearning.AI) · Google Cloud Professional Data Engineer